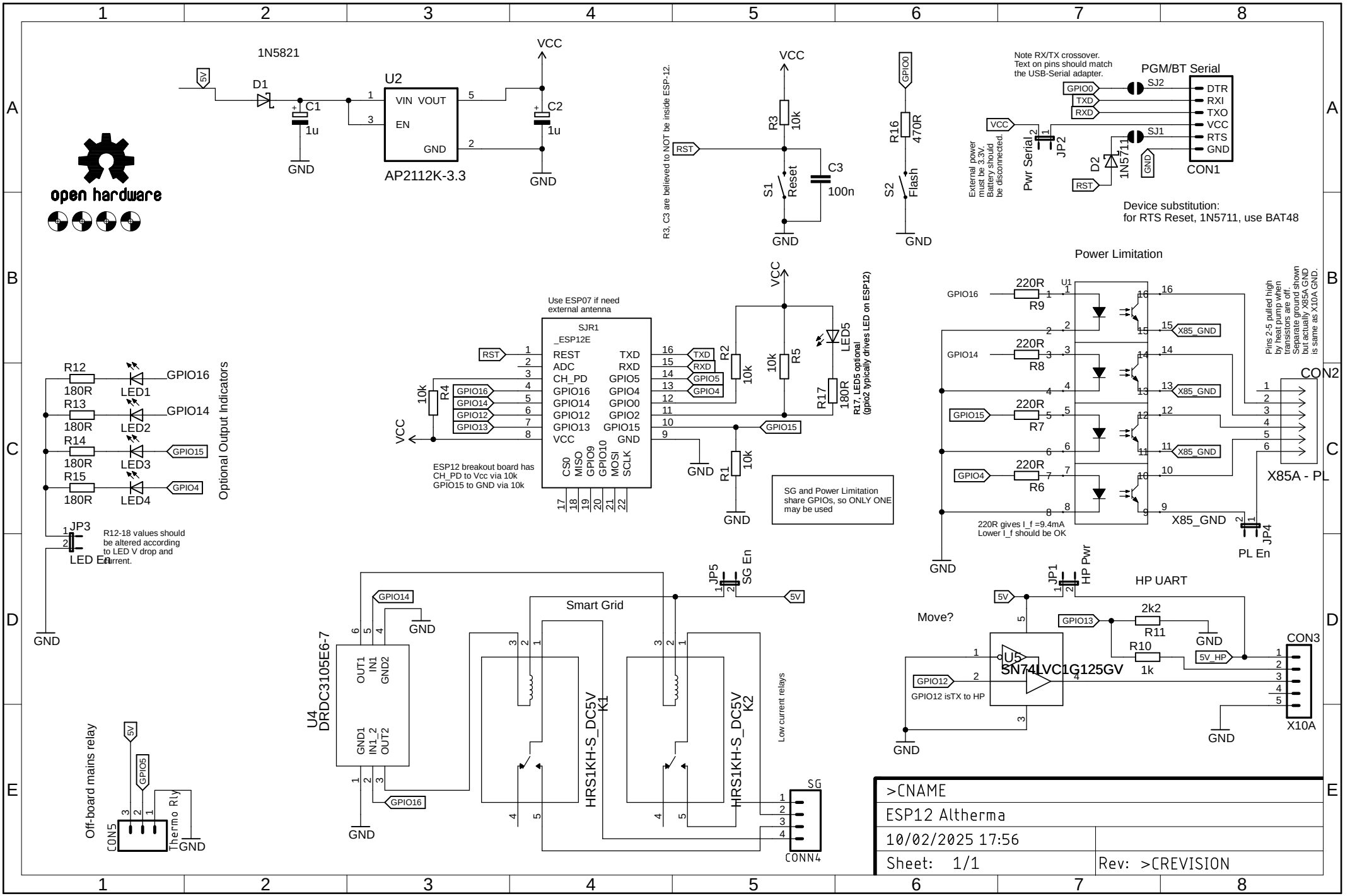
1N5821
D1
C1
1u
U2
VIN VOUT
EN
GND
AP2112K-3.3
C2
1u
VCC
open hardware
R3, C3 are believed to NOT be inside ESP-12.
R3
10k
RST
S1
Reset
C3
100n
R16
470R
S2
Flash
Note RX/TX crossover.
Text on pins should match
the USB-Serial adapter.
PGM/BT Serial
SJ2
DTR
RXI
TXO
VCC
RTS
GND
CON1
SJ1
JP2
Pwr Serial
D2
1N5711
External power
must be 3.3V.
Battery should
be disconnected.
Device substitution:
for RTS Reset, 1N5711, use BAT48
Power Limitation
Use ESP07 if need
external antenna
SJR1
_ESP12E
REST
ADC
CH_PD
GPIO16
GPIO14
GPIO12
GPIO13
VCC
TXD
RXD
GPIO5
GPIO4
GPIO0
GPIO2
GPIO15
GND
CS0
MISO
GPIO9
GPIO10
MOSI
SCLK
R2
10k
R5
10k
R4
10k
R1
10k
LED5
R17
180R
R17, LED5 optional
(gpio2 typically drives LED on ESP12)
ESP12 breakout board has
CH_PD to Vcc via 10k
GPIO15 to GND via 10k
SG and Power Limitation
share GPIOs, so ONLY ONE
may be used
R12
180R
LED1
GPIO16
R13
180R
LED2
GPIO14
R14
180R
LED3
GPIO15
R15
180R
LED4
GPIO4
Optional Output Indicators
JP3
LED En
R12-18 values should
be altered according
to LED V drop and
current.
R9
R8
R7
R6
220R
U1
X85_GND
CON2
X85A - PL
JP4
PL En
Pins 2-5 pulled high
by heat pump when
transistors are off.
Separate ground shown
but actually X85A GND
is same as X10A GND.
220R gives I_f =9.4mA
Lower I_f should be OK
Smart Grid
JP5
SG En
U4
DRDC3105E6-7
OUT1
IN1
GND2
GND1
IN1_2
OUT2
HRS1KH-S_DC5V
K1
K2
Low current relays
SG
CONN4
HP UART
JP1
HP Pwr
Move?
2k2
R11
R10
1k
U5
SN74LVC1G125GV
GPIO12 isTX to HP
5V_HP
CON3
X10A
Off-board mains relay
CON5
Thermo Rly
>CNAME
ESP12 Altherma
10/02/2025 17:56
Sheet: 1/1
Rev: >CREVISION